

Shields

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Shields

Check the figure below: you can see an Arduino Uno board, Voice Box Shield, Wi-Fi shield which will create a Wireless Talking Speak Jet Server.



Shields

I- What is a Shield?

- An Arduino board such as UNO has a general function to control the electric elements that are connected to it. Thus, in some projects, you can add some extra needed features. For example, some projects require internet access, Bluetooth, or motors. As such, a shield will allow you to add the needed elements.
- A shield is an *add-on electronic board* that can be attached to the Arduino Board in order to add an additional function.

Shields

II- Shields from Factory:

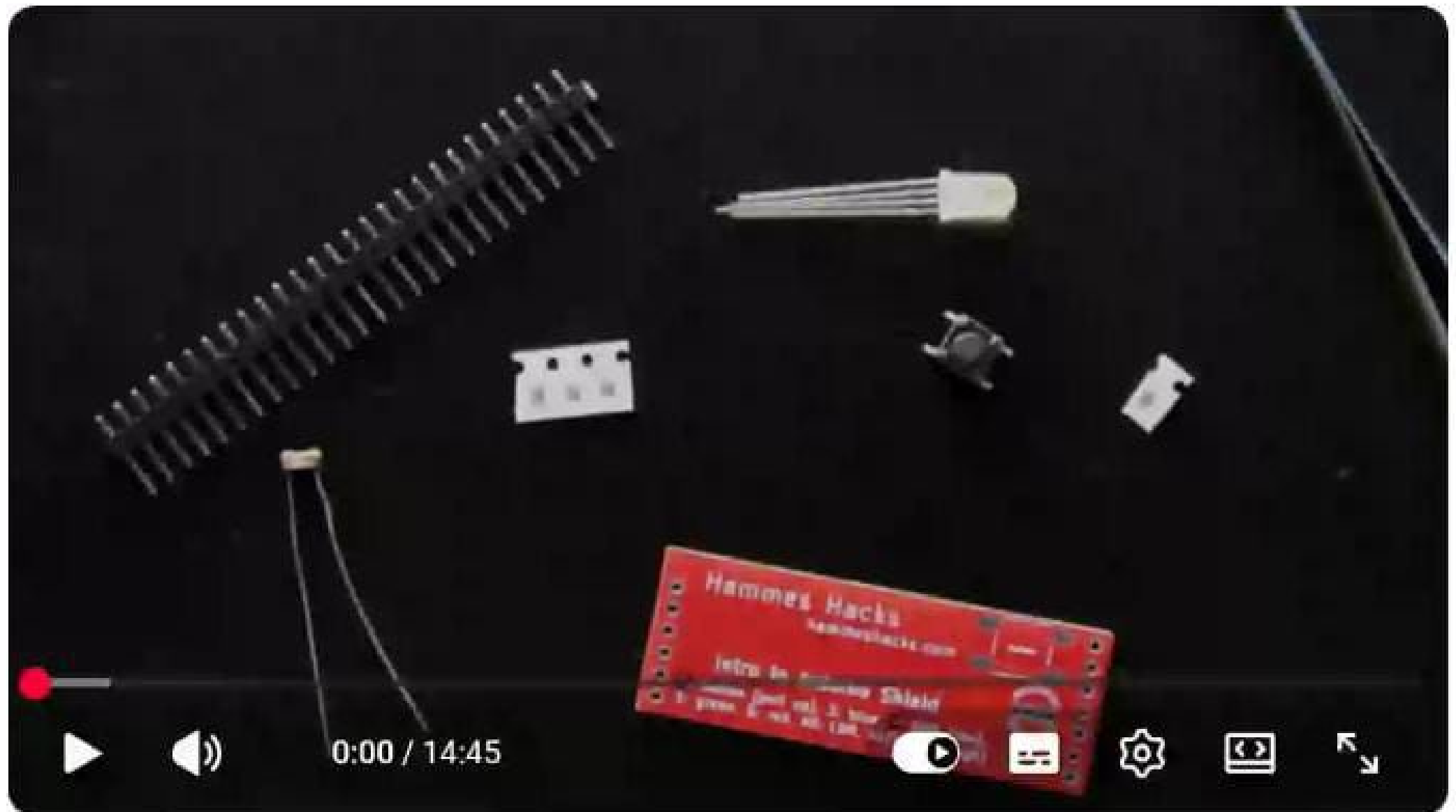
- In order to have all shields connectable to the Arduino, the manufacturer has the shield built with the same shape of the Arduino Uno. This means that a shield should have the same number of pins as an Arduino and at the same position.
- Once you order a shield from a factory, most shields will require soldering. We need to follow some steps to make sure that we have the soldering done correctly.

Shields

1. Pick the correct header :

- The stackable header will allow the stacking of jumper wires on the Arduino shield. Also, it will allow the stacking of another shield if the shields used don't interfere with each other.
- On the other hand, the male header can't support jumping wires or another shield stacked on it. However, you can solder wires directly to the pins. This can be better than the stackable header in some cases.





Intro to Arduino Shield Soldering

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Shields

2. Soldering:

The soldering skill requires practice. In this course, we will not teach you how to solder, but we will mention the steps of soldering. For more information on how to solder you can follow the link below, Soldering process by "Hammes Hacks":

<https://www.youtube.com/watch?v=WlbpW6Rbdml>

a) Insert the Pins: you should have the pins inserted into the shield. Make sure to have them connected correctly before doing any soldering.

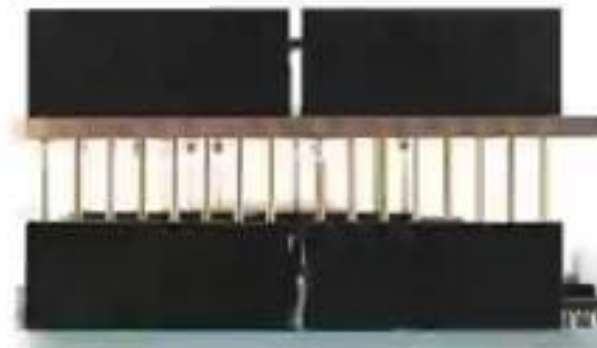


Shields

b) Solder One pin of each header

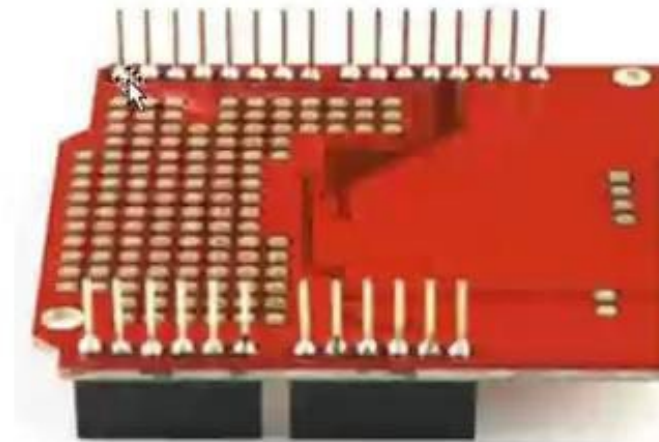


c) Check the header pins direction: it should be perpendicular to the shield.

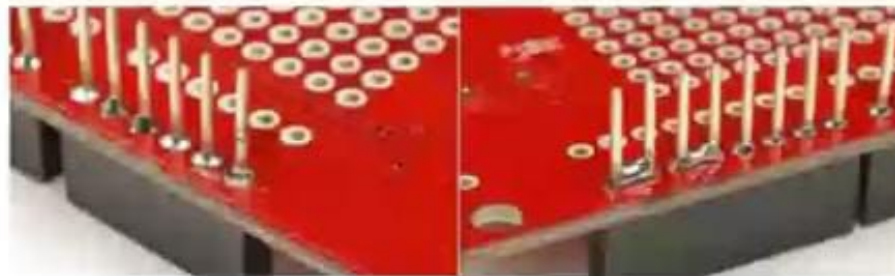


Shields

d) Complete the soldering



e) Check for soldering mistakes: like lost soldering or shorted solder.



Shields

III- Shields Types:

In the market you can find lots of shields which will add an extra function to the Arduino board. We will introduce some of the major shields that can be used with Arduino.

1- ProtoShield Kit:

This shield is of two types: one that has a breadboard attached to it and another one that has free slots for direct soldering.



Shields

2- Joystick Shield: It adds a joystick to your Arduino! The shield sits on top of your Arduino board and turns it into a simple controller. Five momentary push buttons (4+ joystick select button) and a two-axis thumb joystick gives your Arduino functionality on the level of old Nintendo controllers. Soldering is required, but it's relatively easy and requires minimal tools.

3 - MicroSD Shield:

With the microSD shield, you can save the data gathered by your Arduino board. This can be very handy in monitoring applications like irrigation, weather status or home automation.



Shields

4- GPS Shield: This shield will allow your Arduino to track its position, speed, and altitude. The shield can have an SD card connected to it through the USD port, which will facilitate the storing of data.



5- WiFi Shield: This shield will allow the Arduino to be connected to the internet adapter wirelessly. This will equip the Arduino board with internet access. Once the Arduino has internet access, then we can upload data and send data to the Arduino from anywhere.



Shields

IV - How does a shield work?

Most of the shields are equipped with pins that allow the user to directly mount the shield on the Arduino. Once the shield is mounted on the Arduino it will use specific pins in the Arduino to establish a link between the Arduino and the shield. Once the Arduino is turned on, the shield will be powered.



Each shield will use a certain amount of pins to establish a connection with the Arduino.

DC Motor

A DC motor is an electrical element that required Direct current to perform the continuous rotational motion.

The DC motor will rotate once it takes electricity. It will reverse direction if we reverse the electricity flow.

The speed of motor will vary if the voltage changes.
We can control the speed of the motor by varying the duty cycle of the PWM control signal.



DC Motor

Dc motors need high current to operate with the low supply current capabilities of the Arduino boards, there are two methods to control DC motors with Arduino.

Method One:

use a motor shield or motor board that facilitates motor controlling.

I

Method Two:

Build your own H-Bridge circuit using transistors and relays to control the motor.

DC Motor Example

Water pump

Water pump is a good example of DC motor used in Arduino irrigation projects. This is because a pump is a DC motor with a mechanical system that pumps water.



Motor Shield Adafruit

Motor Shield Adafruit



Motor Shield Adafruit

The Adafruit motor shield can control the following

Adafruit Motor shield



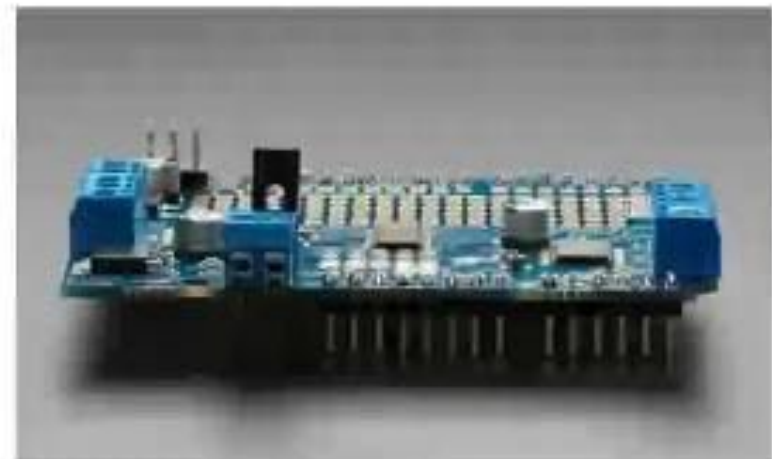
4 DC Motors



2 Stepper Motors

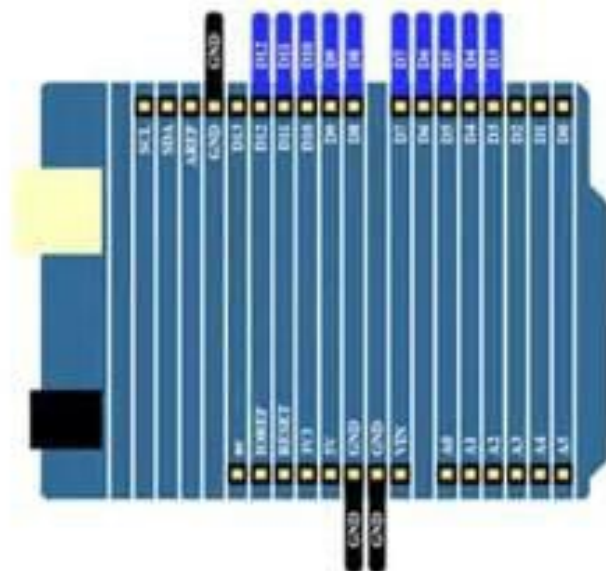


2 Servo Motors



Motor Shield Adafruit

The shield will use specific pins according to the library used. This shield will use the following pins.



Note:

The following pins are in use only if the DC/Stepper noted is in use:

D11: DC Motor #1 / Stepper #1 (activation/speed control)

D3: DC Motor #2 / Stepper #1 (activation/speed control)

D5: DC Motor #3 / Stepper #2 (activation/speed control)

D6: DC Motor #4 / Stepper #2 (activation/speed control)

The following pins are in use if any DC/steppers are used:

D4, D7, D8 and D12 are used to drive the DC/Stepper motors via the 74HC595 serial-to-parallel latch.

The following pins are used only if that particular servo is in use:

D9: Servo #2 control.

D10: Servo #1 control.

Motor Shield Adafruit

The use of this library will facilitate the motor control for both servo and Dc Motors.

`AF_DCMotor motorName(Number);`

This code will define the motor name and the port which has the motor connected to on the motor shield.

Code Terms:

AF_DCMotor : code
motorName: leftMotor
number: from 1 to 4

`motorName.setSpeed(speedValue);`

This code will define the power of which the motor will be receiving. The signal is a PWM ranged between (0 and 255).

motorName: leftMotor
setSpeed: code
speedValue: from 0 to 255

Motor Shield Adafruit

```
Servo1.attach(port  
number);
```

This code will define Arduino port that has the servo linked to. This is related to the default library of the servo motors <servo.h>.

Code Terms:

Servo1: the pin which has the servo on the shield

attach: code

port number: 9 or 10 (this are the pins that support servo motors)

```
servoName.write(i);
```

This code will set the servo motor position. It varies depending on the type of the Servo. Mainly the code uses the degree as a unit.

servoName: arm servo

write code

i: angle of the servo in degrees.

Motor Shield Adafruit

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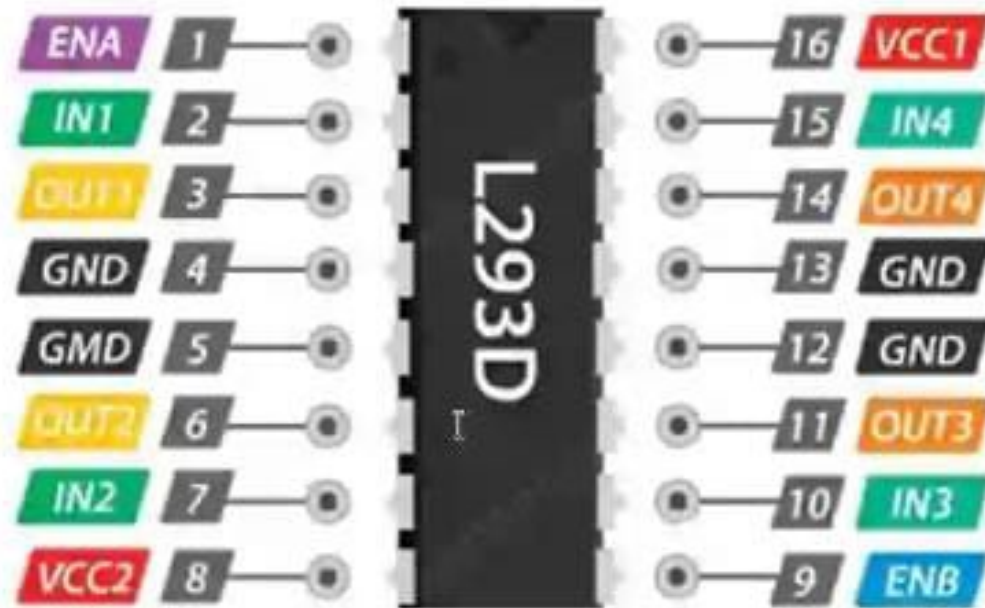
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servoName: arm servo

write code

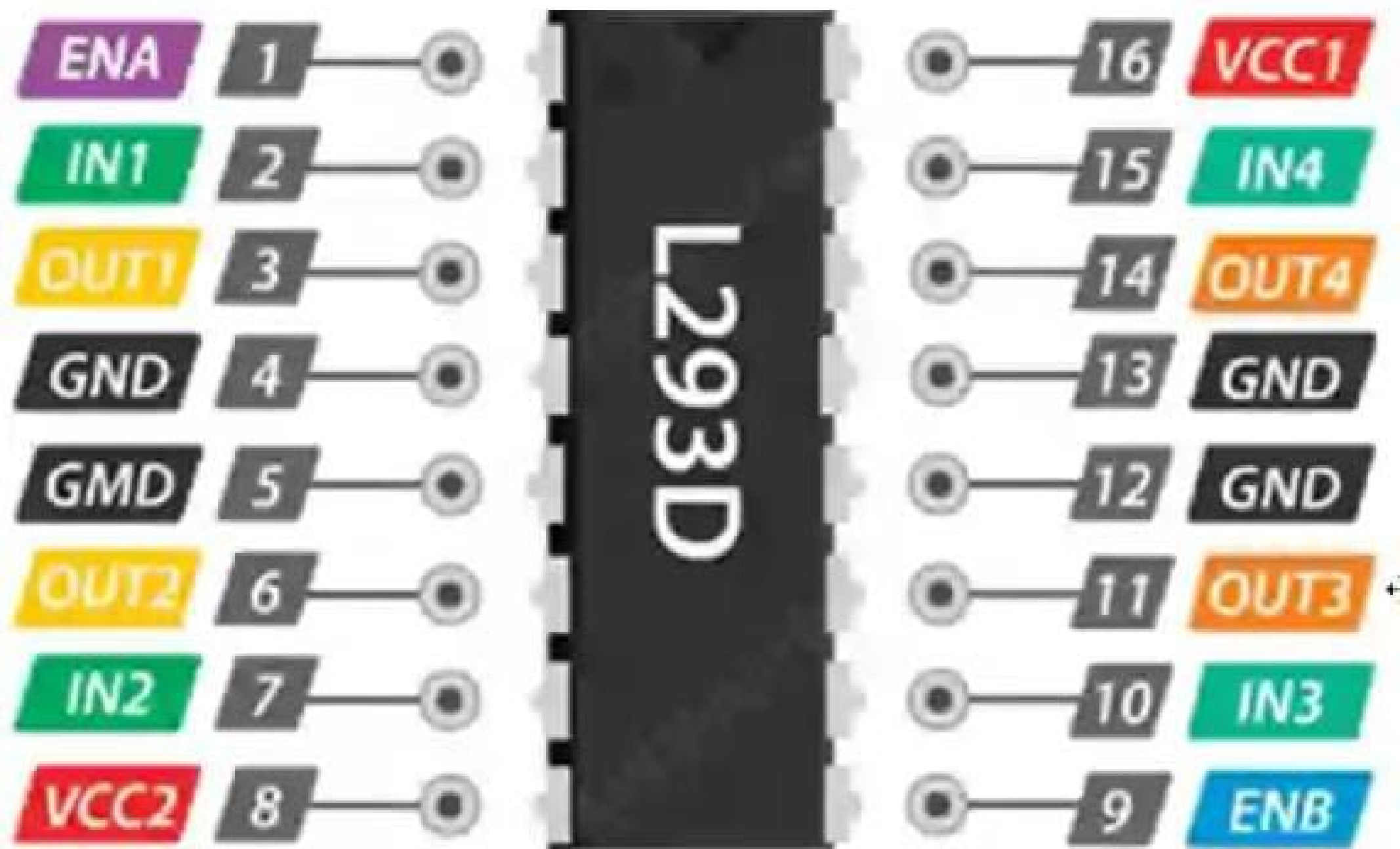
i: angle of the servo in degrees.

H-Bridge

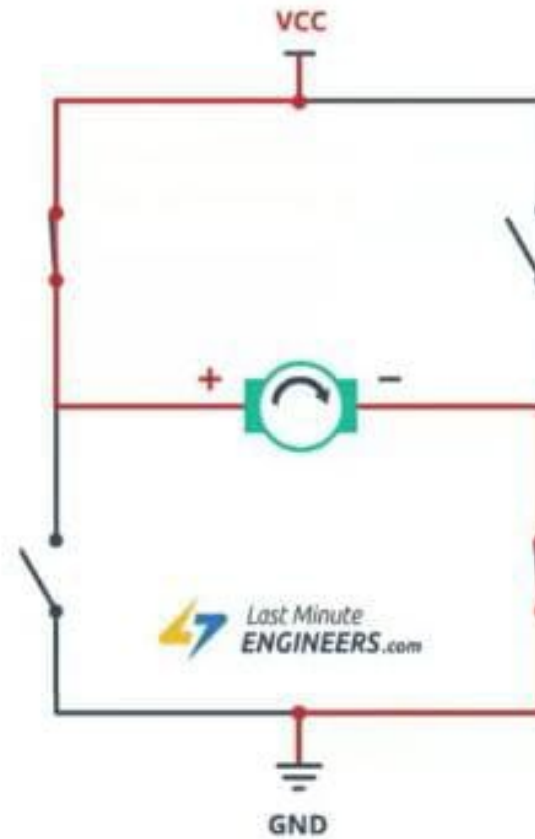
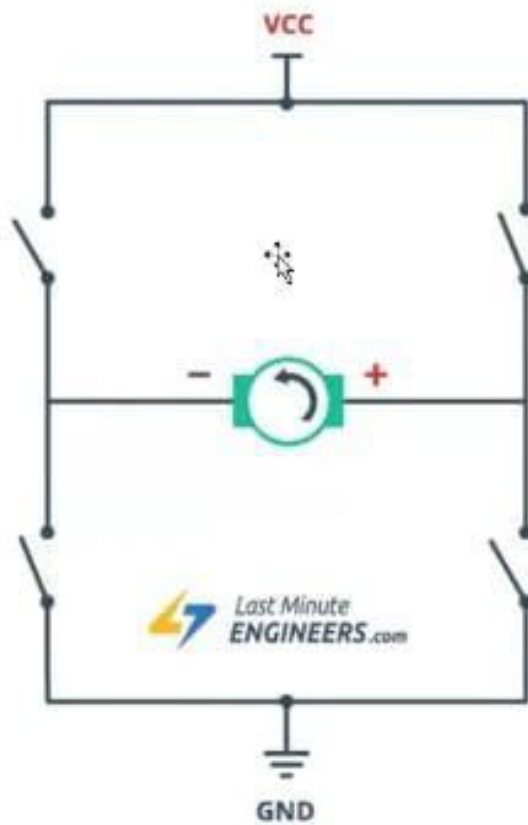


L293D Pinout

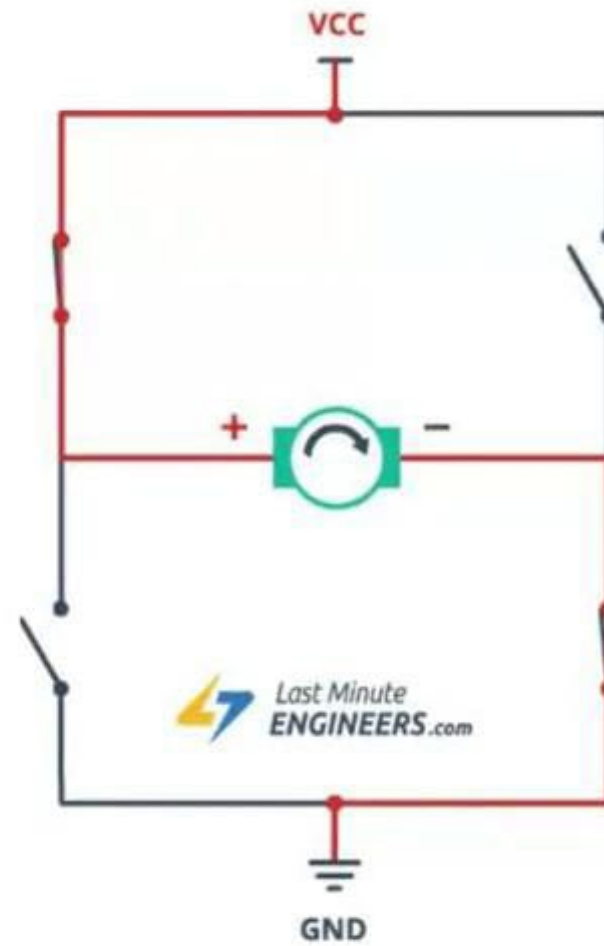
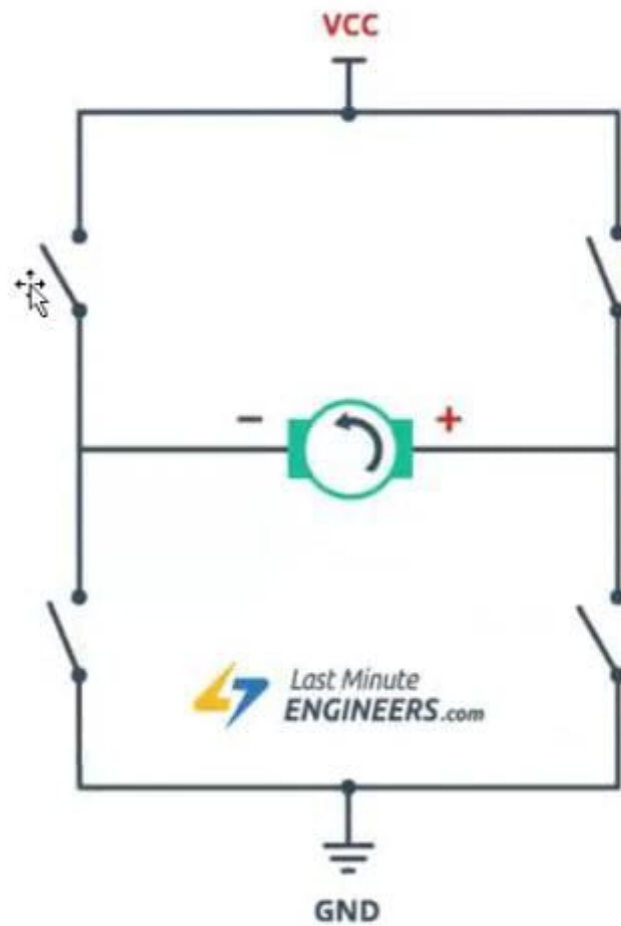
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H-Bridge

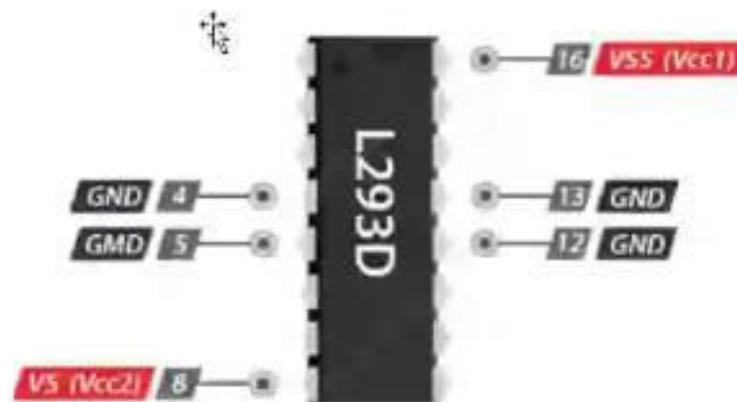


H-Bridge



H-Bridge

Power Pins :



VS (Vcc2) pin gives power to the internal H-Bridge of the IC to drive the motors. You can connect an input voltage anywhere between 4.5 to 36V to this pin.

VSS (Vcc1) is used to drive the internal logic circuitry which should be 5V.

GND pins are common ground pins. All 4 GND pins are internally connected and used to dissipate the heat generated under high load conditions.

H-Bridge

Output Pins

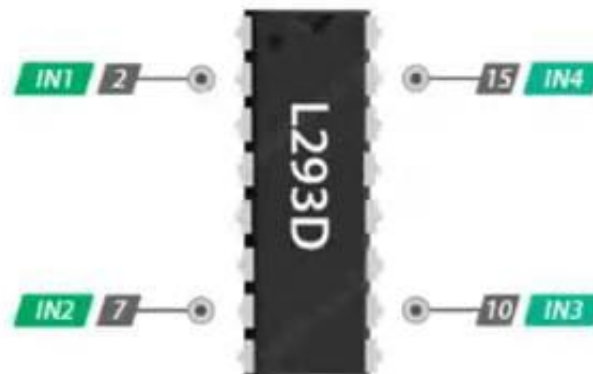


The L293D motor driver's output channels for the motor A and B are brought out to pins **OUT1,OUT2** and **OUT3,OUT4** respectively. You can connect two 5-36V DC motors to these pins.

H-Bridge

Direction Control Pins:

By using the direction control pins, you can control whether the motor rotates forward or backward. These pins actually control the switches of the H-Bridge circuit inside the L293D IC.



The IC has two direction control pins for each channel. The **IN1** and **IN2** pins control the spinning direction of motor A; While **IN3** and **IN4** control the spinning direction of motor B.

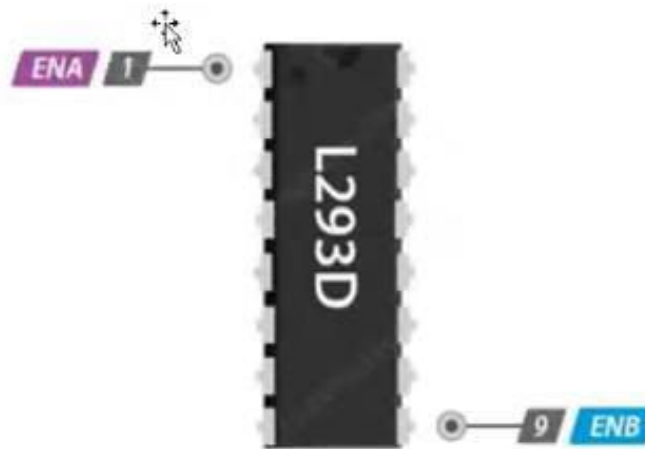
H-Bridge

The spinning direction of the motor can be controlled by applying logic HIGH (5V) or logic LOW (Ground) to these inputs. The chart below shows how this is done.

IN1	IN2	Spinning Direction
Low(0)	Low(0)	Motor OFF
High(1)	Low(0)	Forward
Low(0)	High(1)	Backward
High(1)	High(1)	Motor OFF

H-Bridge

Speed Control Pins



The speed control pins **ENA** and **ENB** are used to turn on/off the motors and control its speed.

Pulling these pins HIGH will cause the motors to spin, while pulling it LOW will stop them. But, with Pulse Width Modulation (PWM), you can actually control the speed of the motors.

Activity 1: Controlling DC Motor

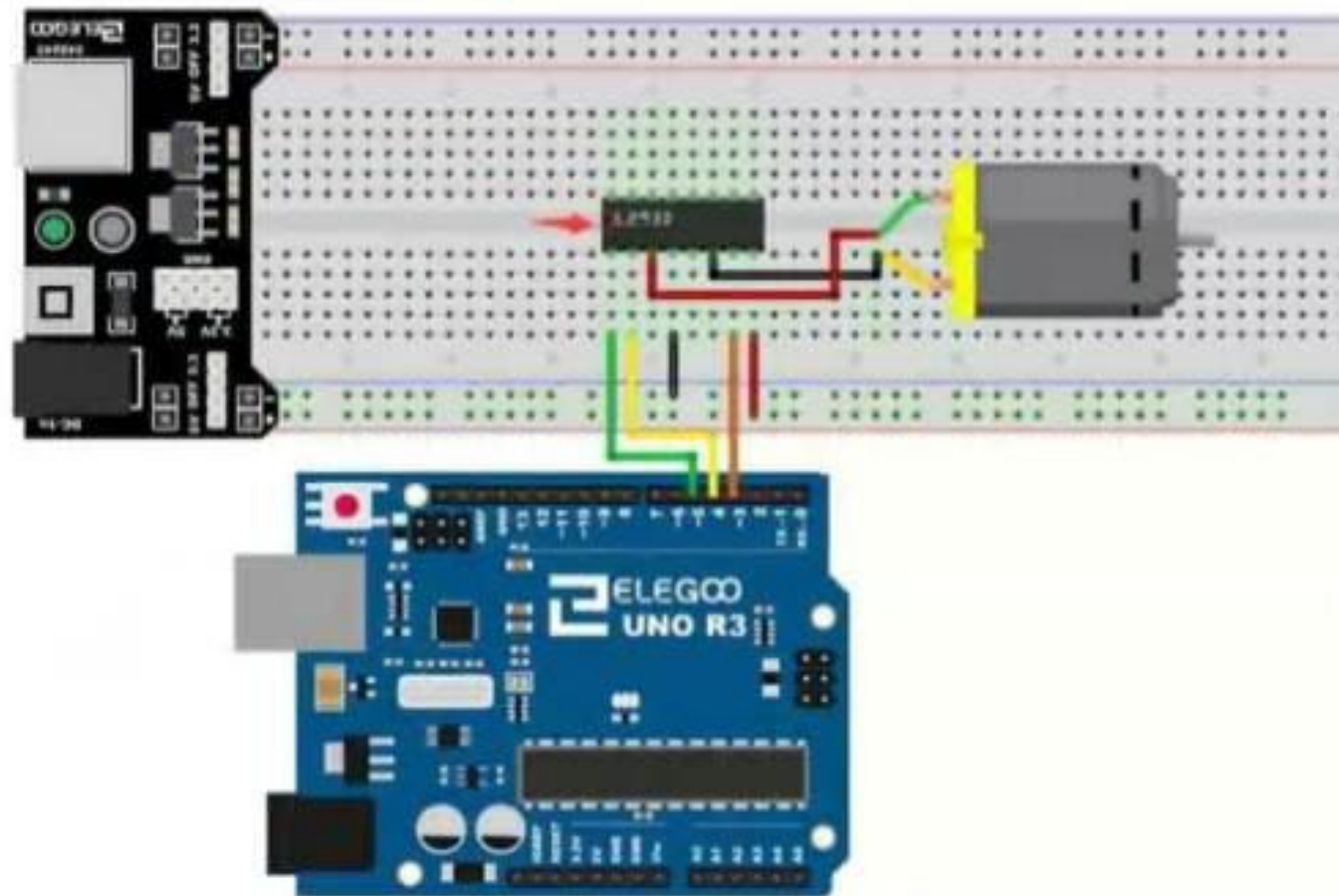
Load speed is constant

Start speed

100%

100%
100%
100%

Board connection



```
File Edit Sketch Tools Help
Arduino UNO
sketch_jul31a.ino
1 int SpeedPin=5; //EnA : pin used to control the speed of the DC Motor using PWM signal
2 int dir1=4; //In1: pin used to control the direction of the DC Motor : Forward
3 int dir2=3; //In2: pin used to control the direction of the DC Motor : Backward
4 int MotorSpeed=150; //constant motor speed (motor starts to spin at a minimum speed of 100RPS)
5 void setup() {
6     // put your setup code here, to run once:
7
8 }
9
10 void loop() {
11     // put your main code here, to run repeatedly:
12
13 }
14
```

```
1 int SpeedPin=5; //EnA : pin used to control the speed of the DC Motor using PWM signal
2 int dir1=4; //In1: pin used to control the direction of the DC Motor : Forward
3 int dir2=3; //In2: pin used to control the direction of the DC Motor : Backward
4 int MotorSpeed=150; //constant motor speed (motor starts to spin at a minimum speed of 100RPS)
5 int redled=8;
6 int blueled=9;
7
8 void setup() {
9   // put your setup code here, to run once:
10  pinMode(SpeedPin,OUTPUT);
11  pinMode(dir1,OUTPUT);
12  pinMode(dir2,OUTPUT);
13  pinMode(redled,OUTPUT);
14  pinMode(blueled,OUTPUT);
15 }
16
17 void loop() {
18   // put your main code here, to run repeatedly:
19   ↑
20 }
21
```

sketch_jul31a.ino

```
4 int MotorSpeed=150; //constant motor speed (motor starts to spin at a minimum speed of 100RPS)
5 int redled=8;
6 int bluled=9;
7
8 void setup() {
9     // put your setup code here, to run once:
10    pinMode(SpeedPin,OUTPUT);
11    pinMode(dir1,OUTPUT);
12    pinMode(dir2,OUTPUT);
13    pinMode(redled,OUTPUT);
14    pinMode(blueled,OUTPUT);
15 }
16
17 void loop() {
18     // put your main code here, to run repeatedly:
19
20     //Turn on the motor with controlling spinning direction 1 : backward or clockwise
21     digitalWrite(dir1,HIGH);
22     digitalWrite(dir2,LOW);
23     analogWrite(SpeedPin,MotorSpeed);
24     digitalWrite(blueled,HIGH);
25 }
```

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